Operating Systems & Networks
CTEC1704C

Lecture – 9 : Introduction to the Internet Protocol Suite

Dr. Sameer Jha, SFHEA
Faculty of Computing
De Montfort University Cambodia

Outline

- The Internet and its structure
- Protocols
- Internet protocol stack
- Physical media

References

- https://www.ietf.org/rfc.html
- https://en.wikipedia.org/wiki/Internet_Protocol_Suite
- https://en.wikipedia.org/wiki/Osi_7_layer_model

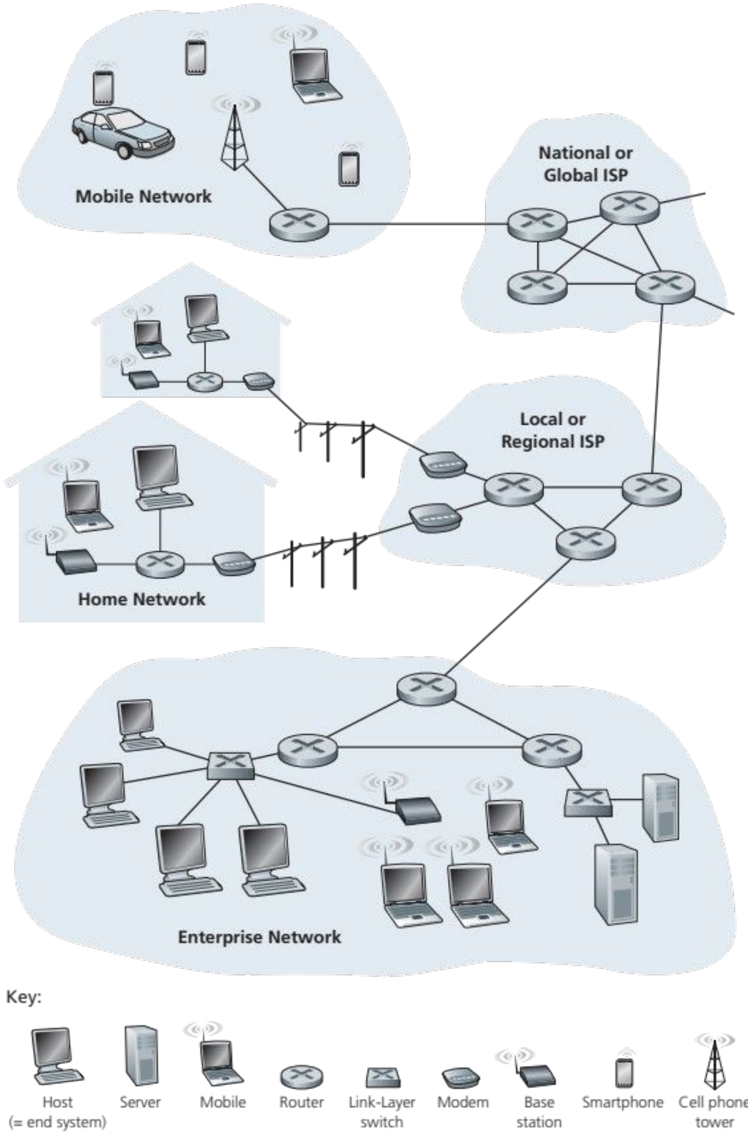
Post lecture discussion topics

- Read the original research paper that described what became the Internet:
  - https://www.cs.princeton.edu/courses/archive/fall06/cos561/papers/cerf74.pdf
- Read https://en.wikipedia.org/wiki/Communication_protocol
  - especially the section headed Protocol Development
- From memory, list the 5 layers of the TCP/IP protocol stack, and give a brief explanation of the role that each layer plays in sending and receiving an email

Lecture Content

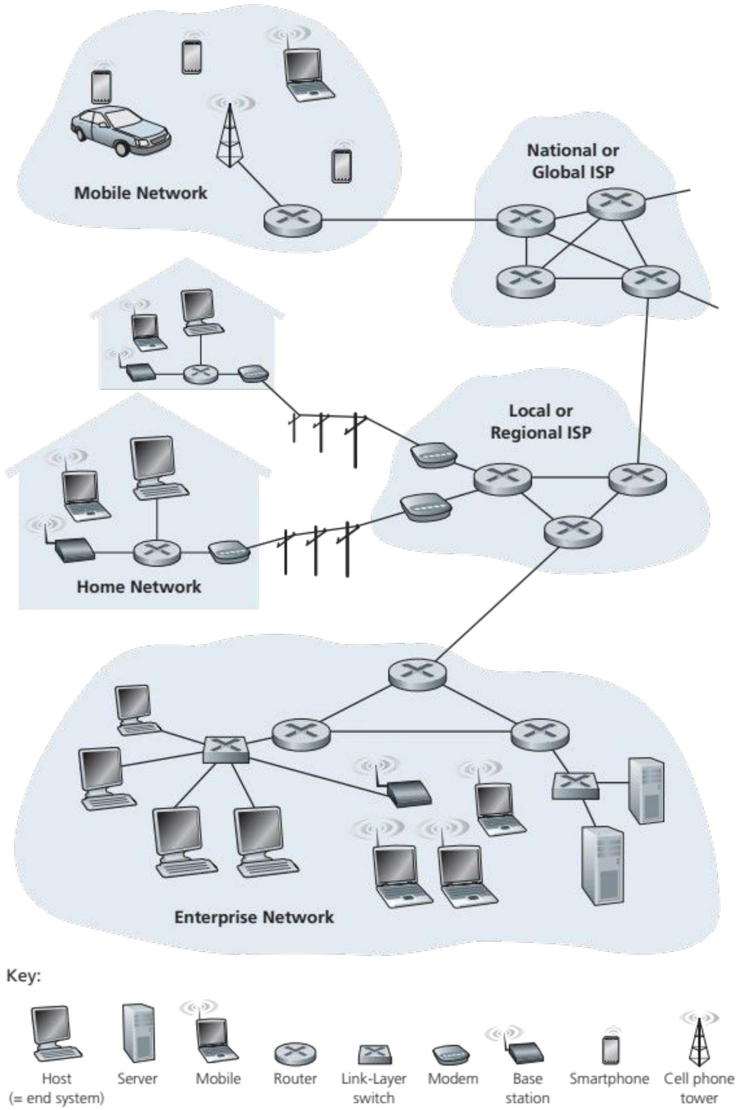
1. **The Internet and its structure**
2. Protocols
3. Internet protocol stack
4. Physical media

The Internet



- The internet connects millions of **devices** around the world
- Devices are called **hosts** or end systems
- These are connected with
  - communication links
  - packet switches

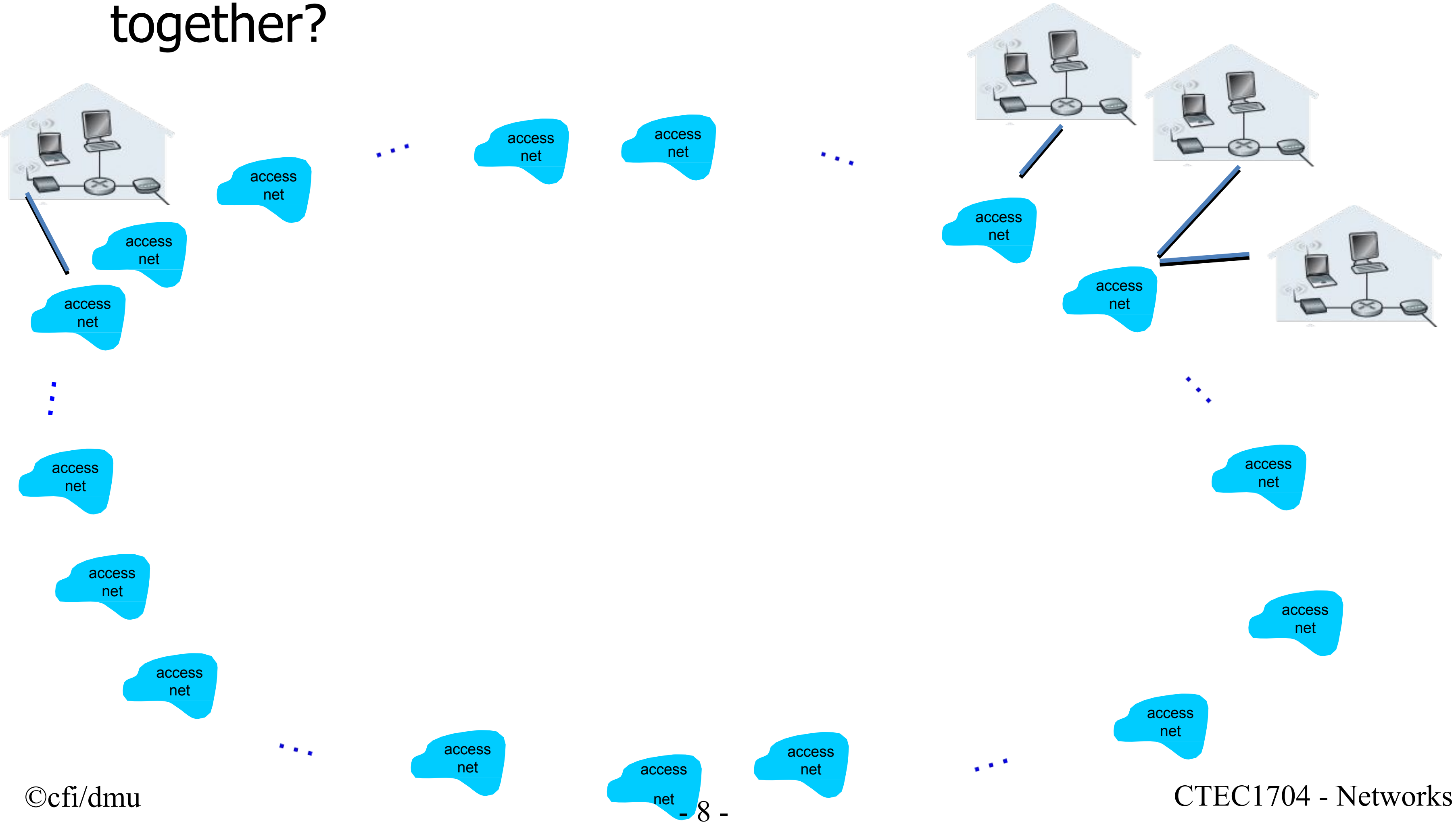
The Internet



- To send information the system segments data into **packets**
  - header information included for each packet that is sent over the network
- Packets are directed through the network to their destination using **routers** and link-layer **switches**

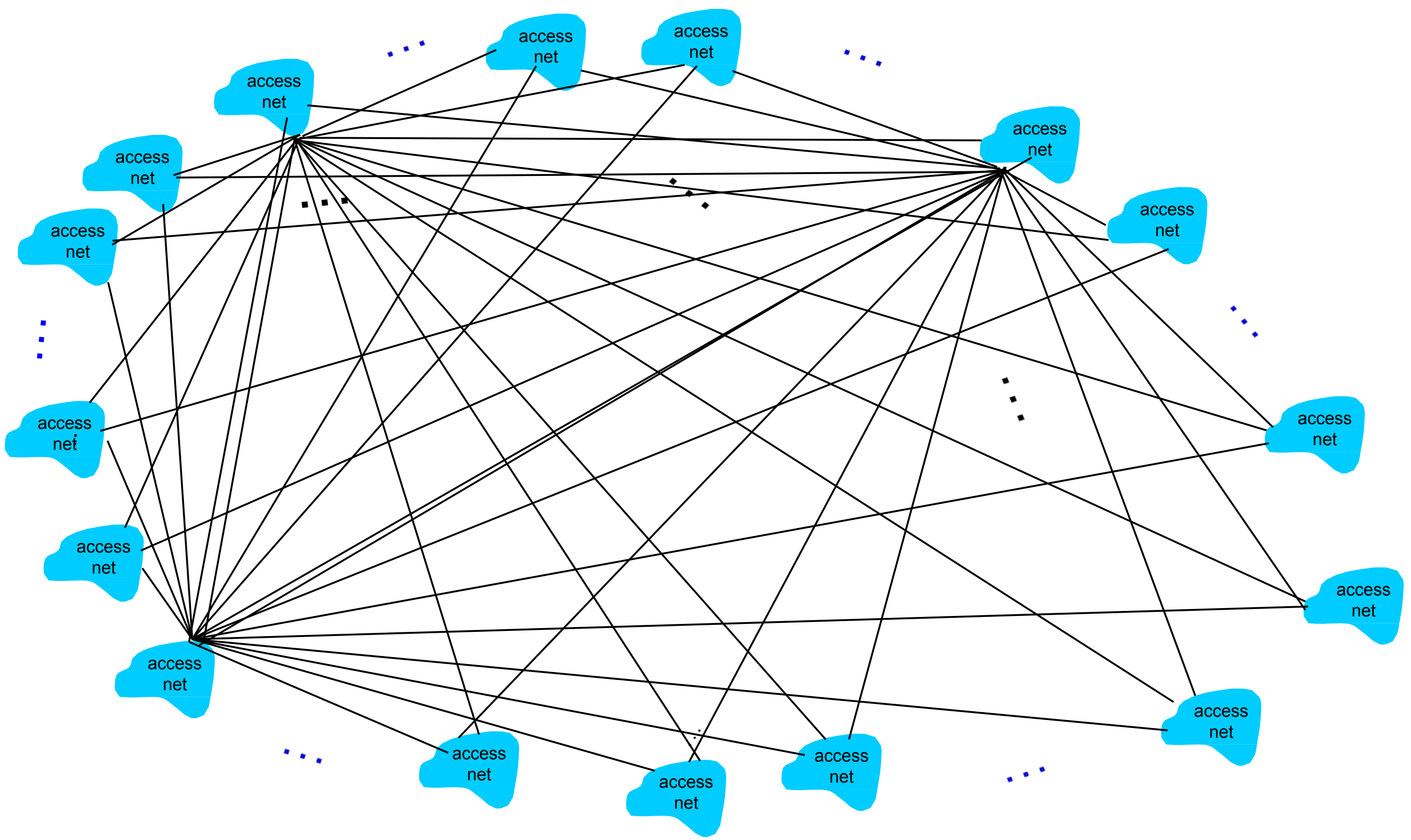
Structure of the Internet

- Given millions of ISPs, how to connect them together?



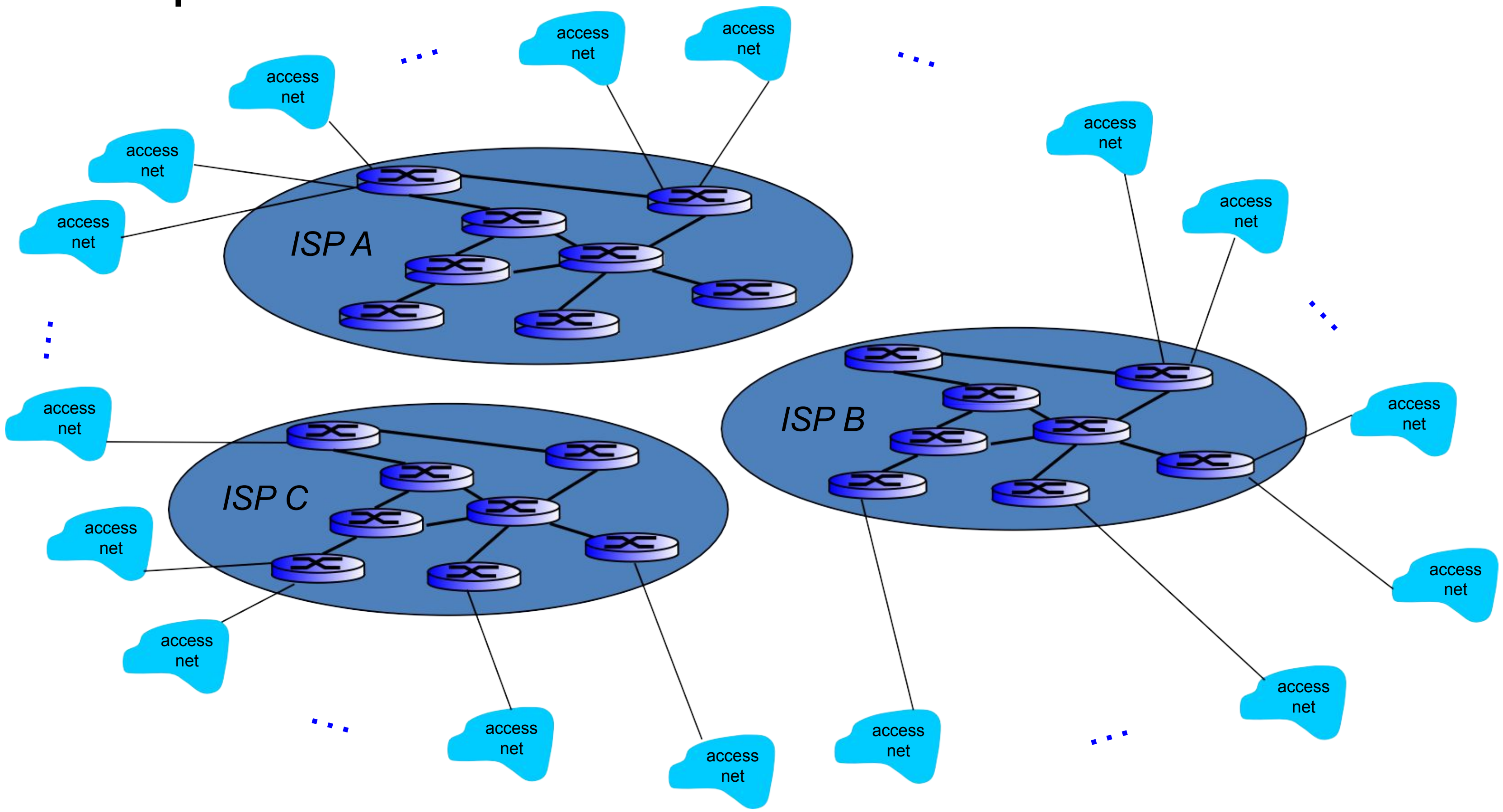
Structure of the Internet

- Option: connect each ISP to every other ISP



Structure of the Internet

- If global ISP is viable business, there will be competitors

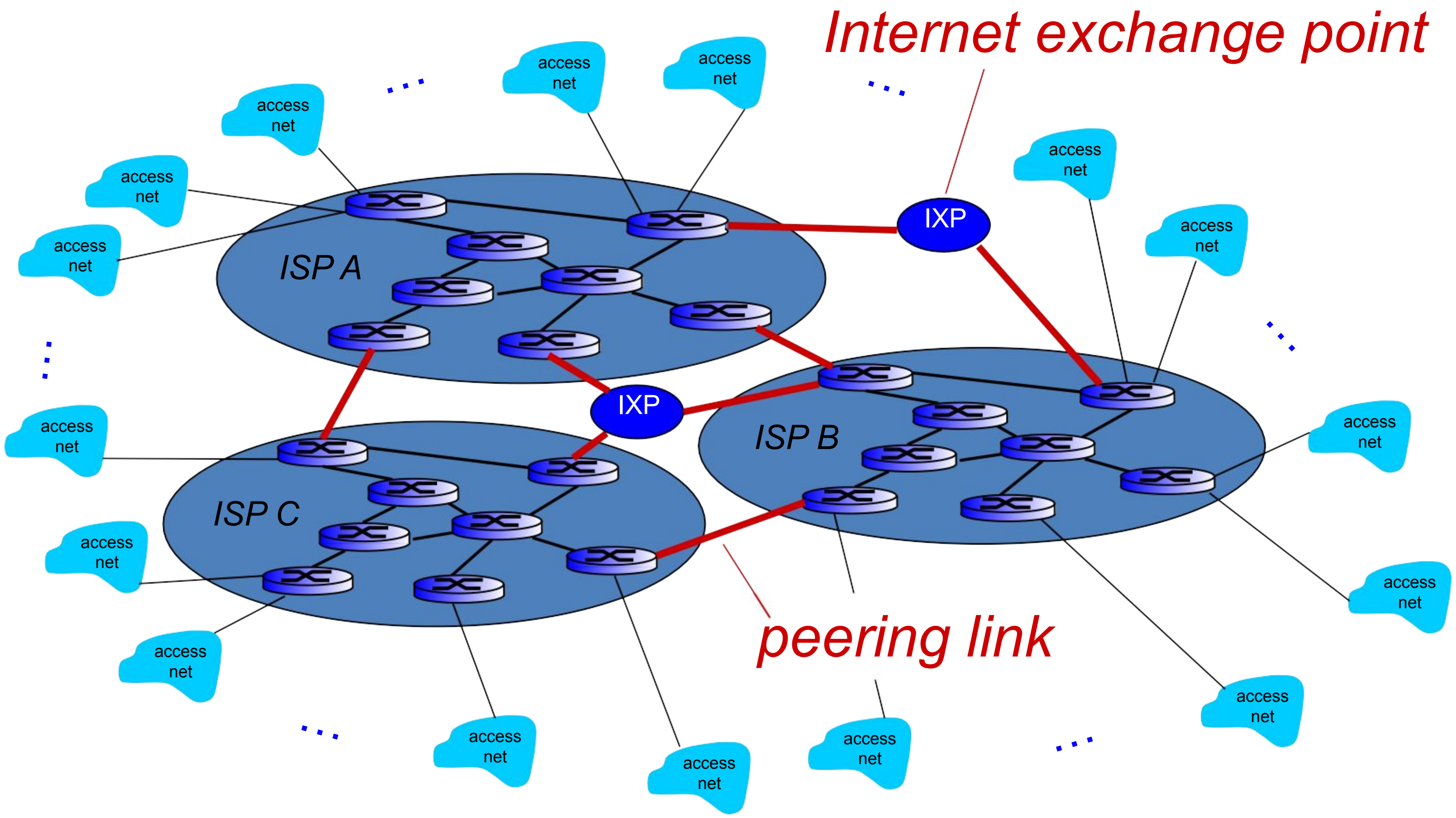


Global ISPs

- Today: between 10 and 20 global ISPs
- For example
  - AT&T (AS7018)
  - Level 3 (AS3356, AS3549)
  - Deutsche Telekom (AS3320)
- Autonomous systems (AS) are entities that participate in global routing in the Internet
- Typically Internet service providers (labelled “access net” in diagrams)

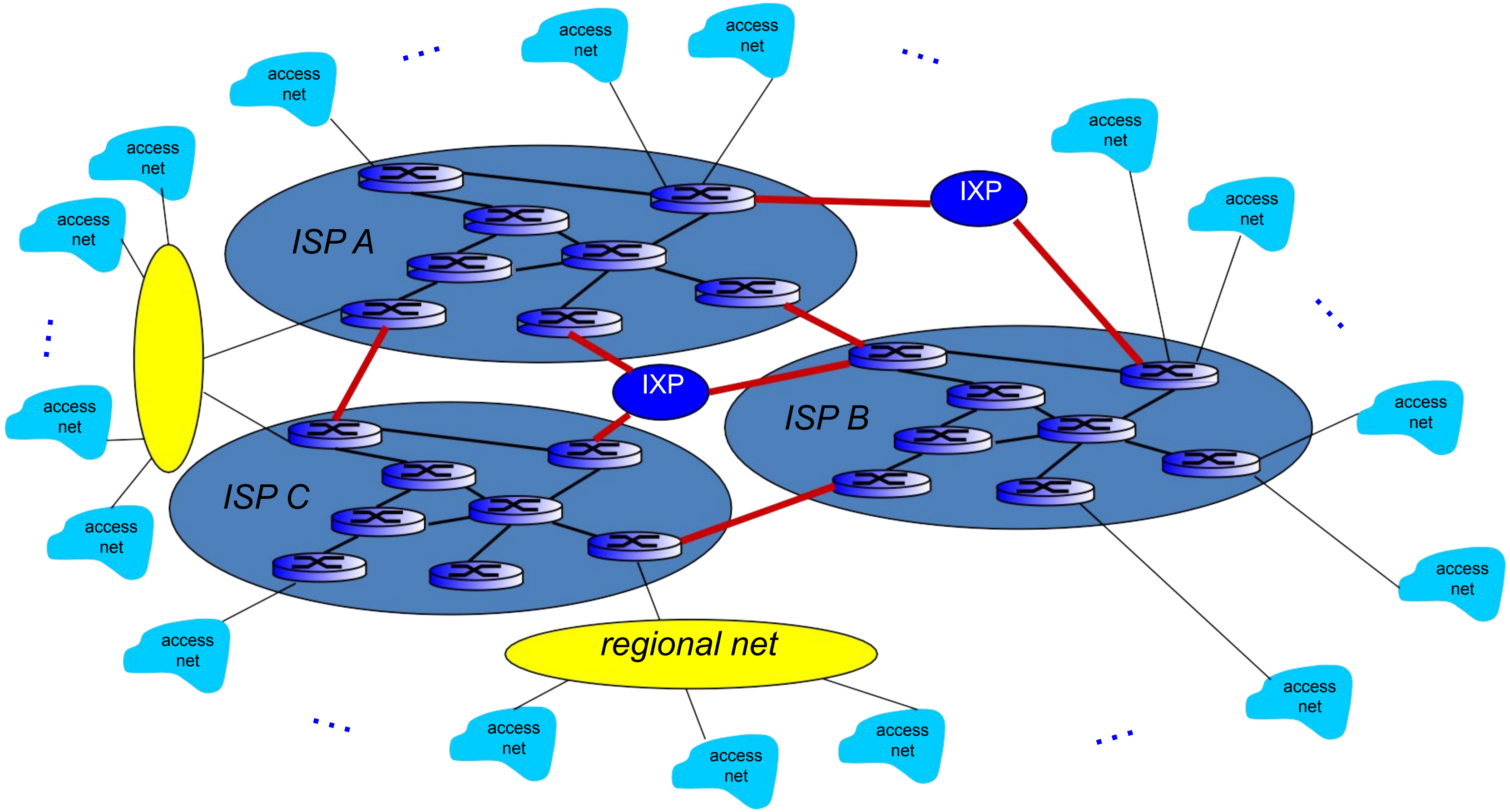
Structure of the Internet

- Have to interconnect all the global ISPs



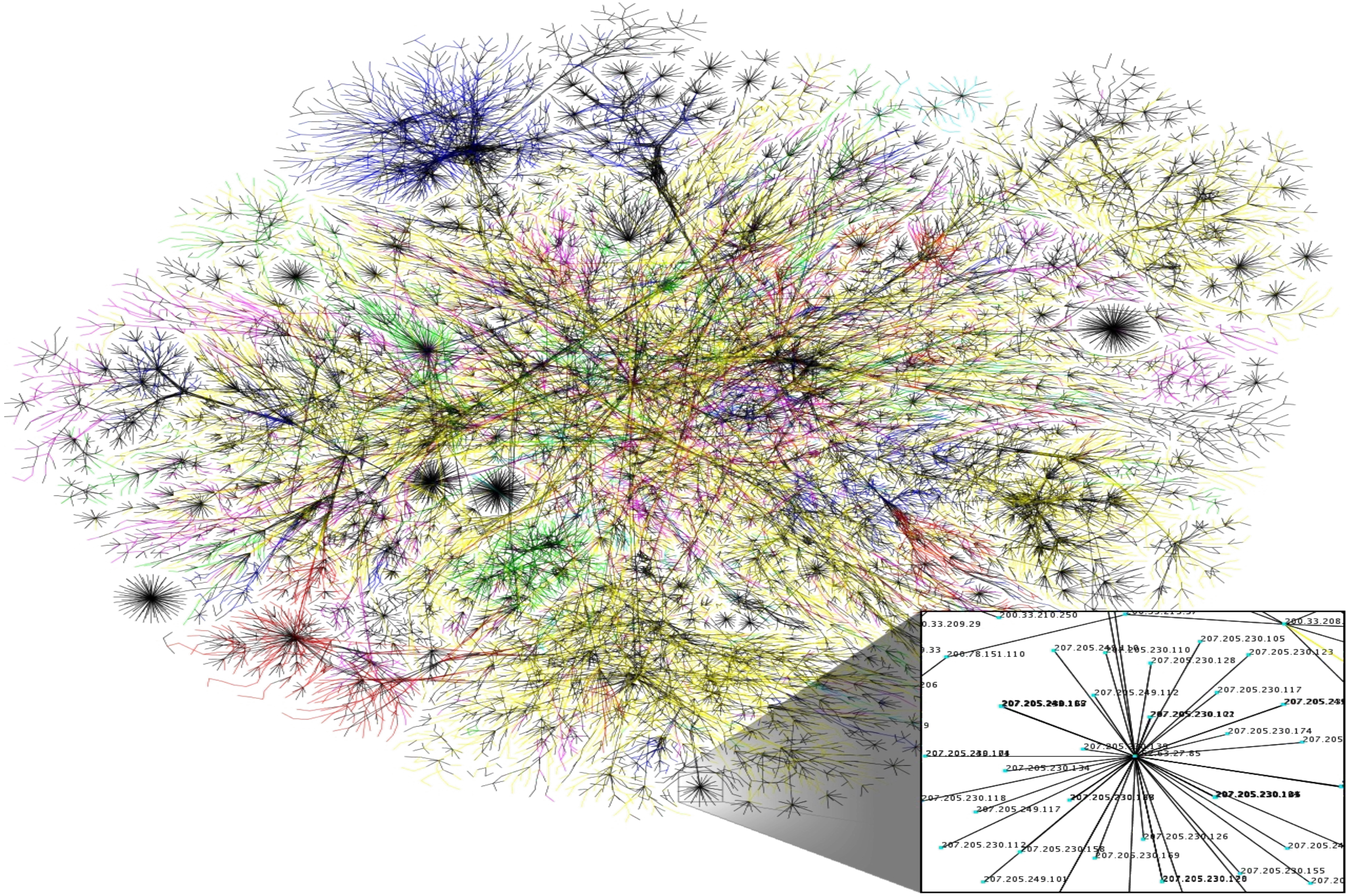
Structure of the Internet

- Regional networks connect local to global ISPs



Regional Networks in the UK

- British Telecom/AS5400
  - buys transit from NTT Communications Corp/AS2914 and Level 3/AS3356
- KCOM Group/AS12390
  - buys transit from KPN/AS286 and Level 3/AS3356
- Vodafone AS3209
  - buys transit from Level3/AS3356, GBLX/AS3549, TeliaSonera International Carrier/AS1299, and Interoute/AS8928

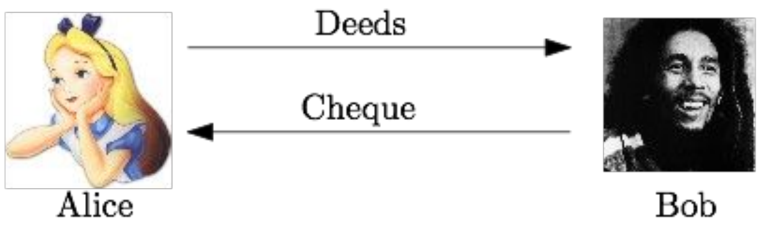

200.33.210.250
207.205.230.105
207.205.230.128
207.205.230.123
200.78.151.110
207.205.249.112
207.205.230.117
207.205.230.174
207.205.230.134
207.205.230.118
207.205.249.117
207.205.230.126
207.205.230.155
207.205.249.101

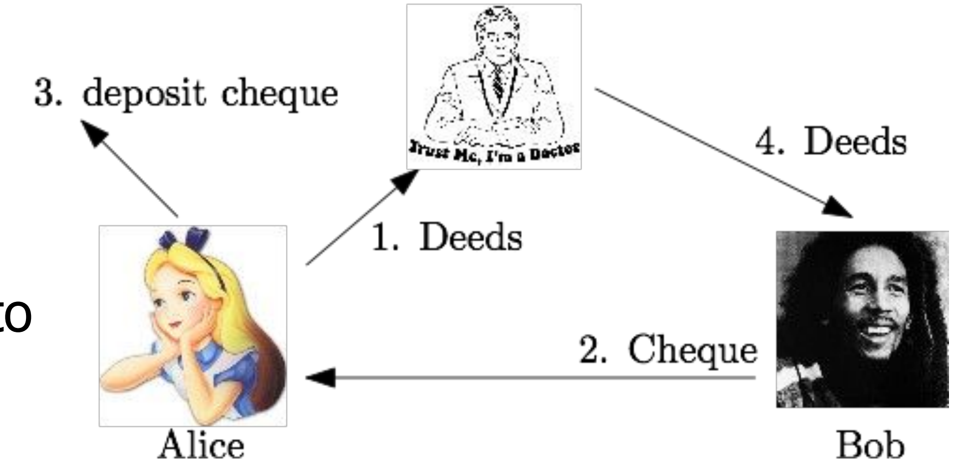
Lecture Content

1. The Internet and its structure
2. **Protocols**
3. Internet protocol stack
4. Physical media

Protocols

- Selling your car is an example of a simple protocol
- Sometimes protocols involve more than two parties
  - if not all information is known (eg DNS)
  - if the purpose cannot be fulfilled at the selected endpoint (eg mail)
  - to arbitrate a protocol to overcome issues of trust



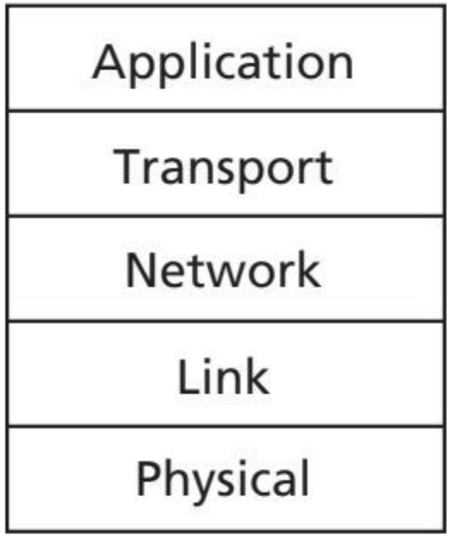


Lecture Content

1. The Internet and its structure
2. Protocols
3. **Internet protocol stack**
4. Physical media

Protocols in the Internet

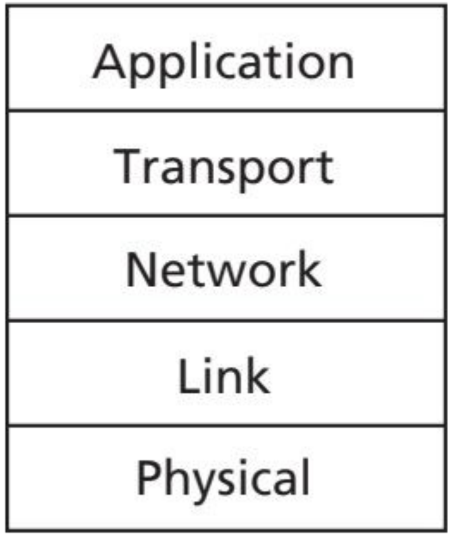
- **Application Layer**
- Contains protocols such as
  - HTTP (web browsing)
  - SMTP (email)
  - FTP (file transfer)
  - DNS (domain name service)
- At this level packets sent are called messages



a. Five-layer Internet protocol stack

Protocols in the Internet

- **Transport Layer**
- Purpose is to transport messages between end points
- Two main transport protocols are TCP (connection) and UDP (connectionless)
- TCP guarantees delivery and flow controls
- At this level packets are called segments

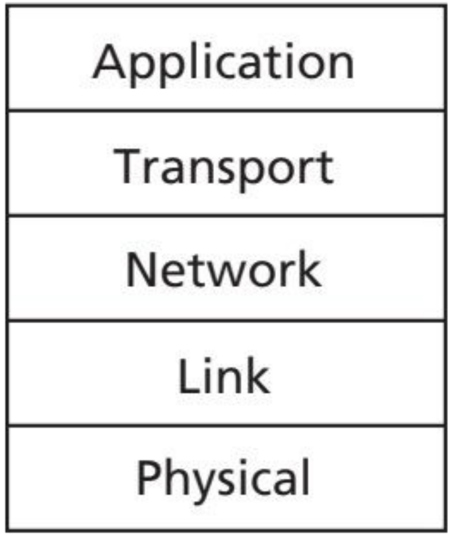


a. Five-layer Internet protocol stack

Protocols in the Internet

- **Network Layer**

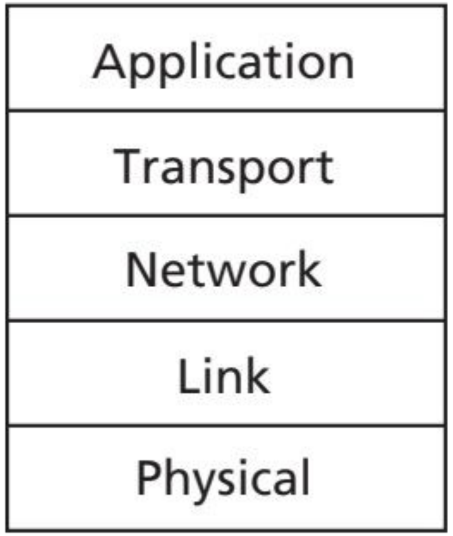
- Provides delivery service
- Uses the Internet protocol (IP) to determine addresses
- Routing protocols at this level determine the path the datagrams (network-layer packets) follow



a. Five-layer Internet protocol stack

Protocols in the Internet

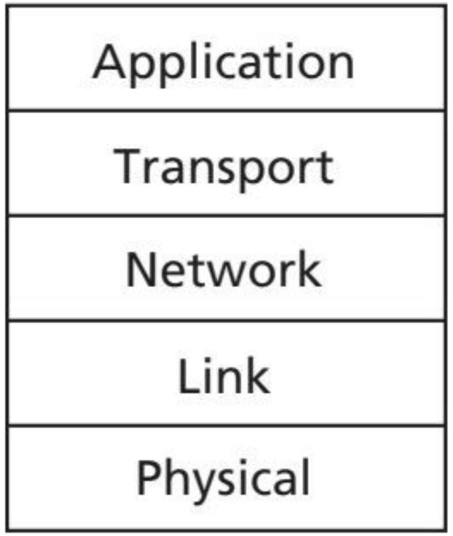
- **Link Layer**
- Delivers packets from node to node in the network
- These are called frames
- Examples of protocols are Ethernet or PPP
- Protocols may differ from link to link (hop)



a. Five-layer Internet protocol stack

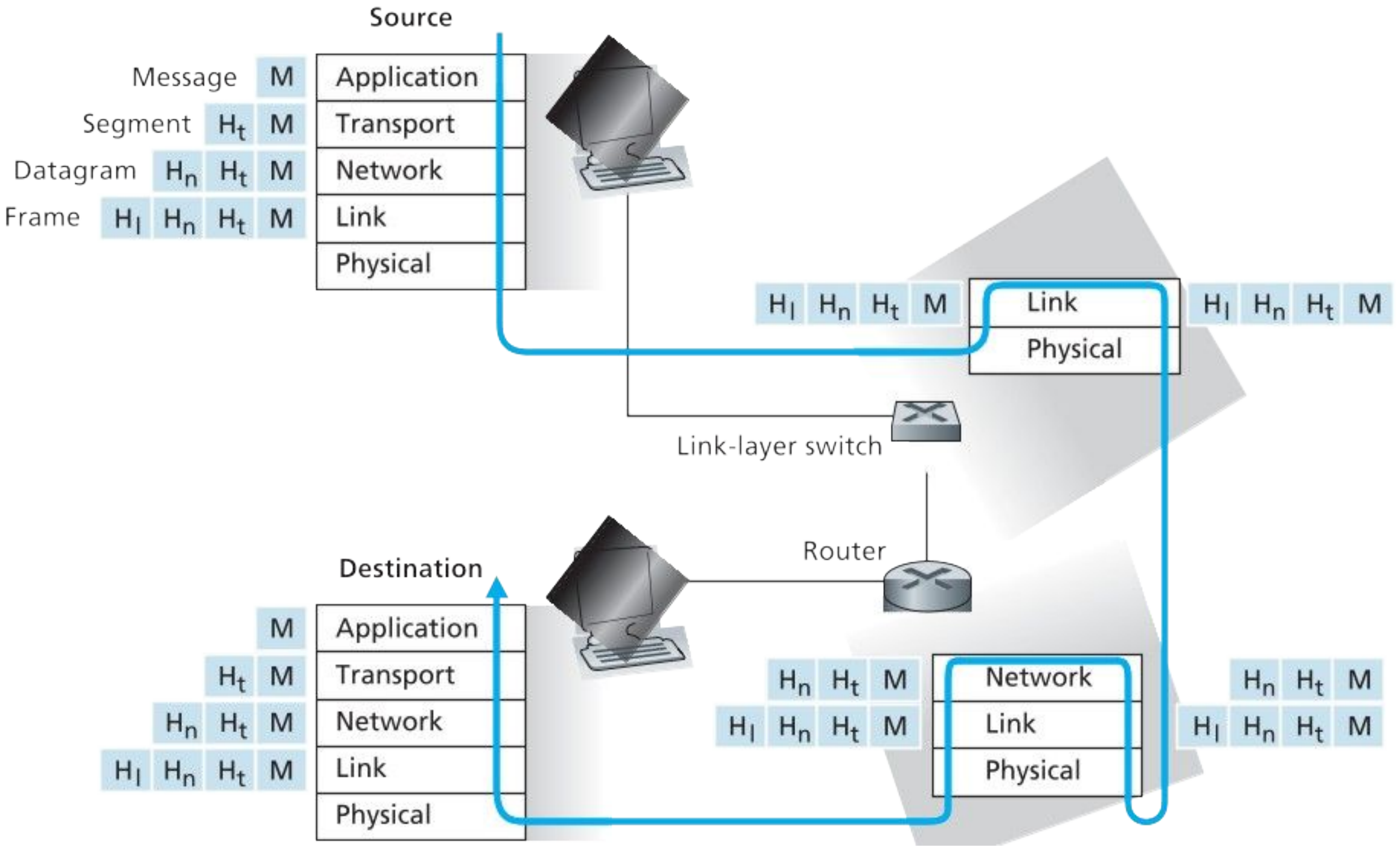
Protocols in the Internet

- **Physical Layer**
- Move individual bits from node to node



a. Five-layer Internet protocol stack

Packets traversing a network



Example

- Let's say we open a website in our browser
- What does the packet look like that we receive from the server?

Application layer					HTTP header	Message
Transport layer				TCP header	HTTP header	Message
Network layer			IP header	TCP header	HTTP header	Message
Link layer		Ethernet header	IP header	TCP header	HTTP header	Message
Physical layer	(encode packet as physical characteristics that represent bits)					

Lecture Content

1. The Internet and its structure
2. Protocols
3. Internet protocol stack
4. **Physical media**

Physical Media

- We will skip protocols on the physical layer
- Physical media: the technology that carries bits from node to node

Physical Media

- Guided media
  - signals propagate in solid media
    - eg copper, fibre, co-ax
- Unguided media
  - signals propagate freely
    - eg radio, microwave

Lecture Content Summary

1. The Internet and its structure
2. Protocols
3. Internet protocol stack
4. Physical media

DE MONTFORT UNIVERSITY CAMBODIA

DE MONTFORT UNIVERSITY CAMBODIA

Thank You!